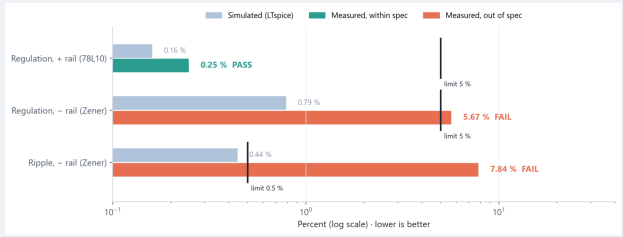
COURSE LABS · 2025–2026

Four labs, one slide each

The question, one graph, the numbers, and what I concluded. Graphs rebuilt from my submitted reports' data; the workbook behind them recalculates every result.

Anmol Bhandohal

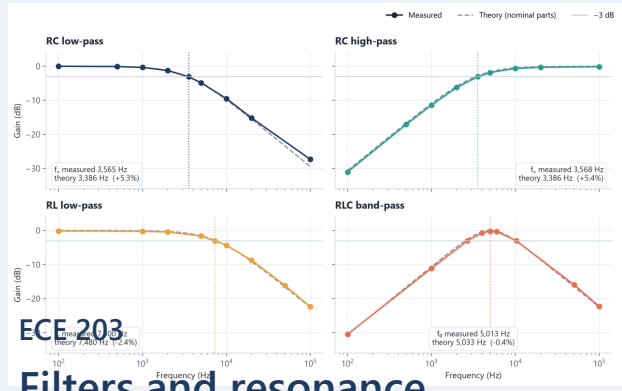
BSc Electrical Engineering (Co-op) · University of Alberta



ECE 302

±10 V dual-rail power supply

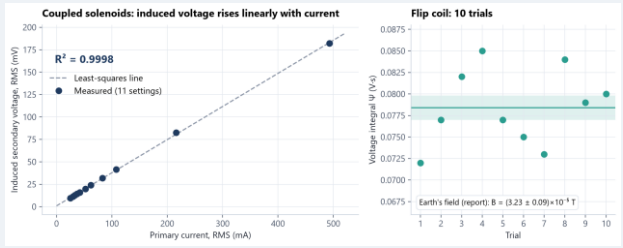
Design, simulate in LTspice, build, measure



ECE 203

Filters and resonance

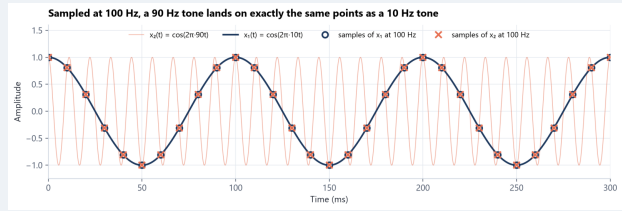
Four filters swept 100 Hz to 100 kHz



PHYS 230

Faraday's law

Earth's field from a flip coil, with error propagation



ECE 340

Sampling and aliasing

MATLAB: when 90 Hz becomes 10 Hz

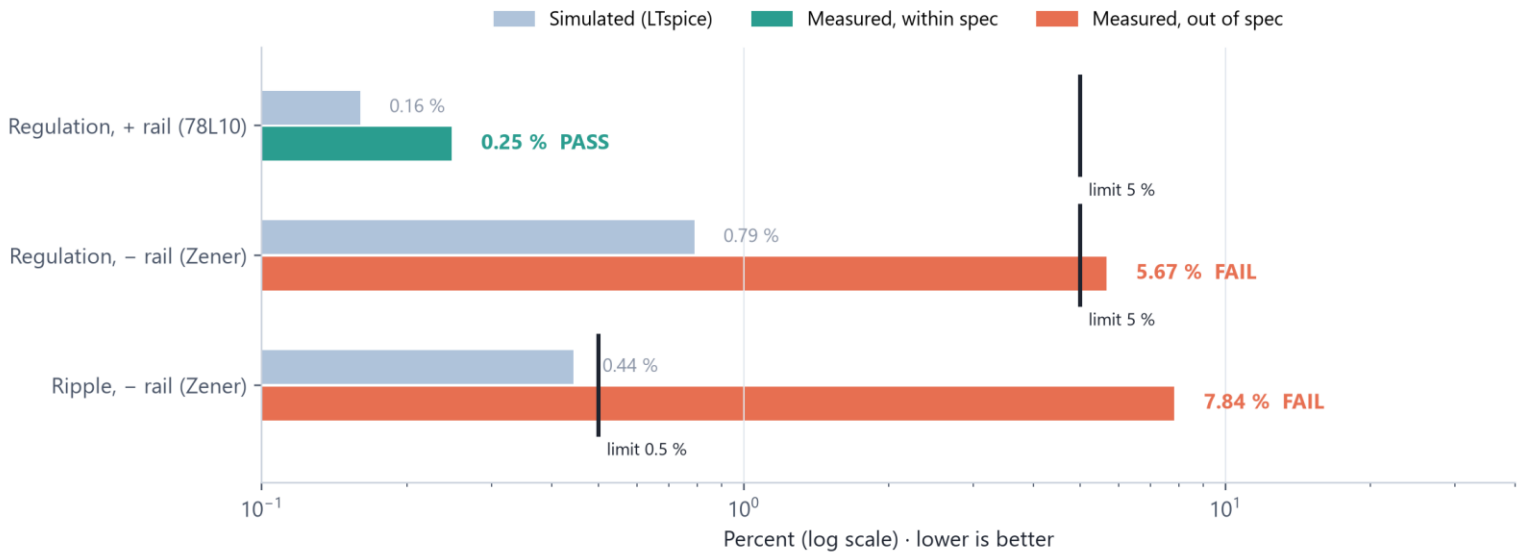
A ±10 V supply: one rail passed, one didn't, and why

THE QUESTION

Can a transformer, rectifiers, a 78L10 regulator and a Zener shunt hold ±10 V within 5 % regulation and 0.5 % ripple at full load?

WHAT WE DID

Sized the 220 Ω ballast by hand, simulated both rails in LTspice, built it on a breadboard and measured with a scope and meter.



CONCLUSION

The IC regulator rail met spec. The Zener rail fell out of breakdown under heavy load: not enough ballast current. Fix: more filter capacitance or a smaller ballast.

0.25 %
+ rail regulation, measured (limit 5 %)

5.67 %
− rail regulation, measured: over the limit

7.84 %
− rail ripple, measured (sim 0.44 %, limit 0.5 %)

Group lab with Rohan Singh. Values from our submitted report (Oct 2026); chart rebuilt from them.

Four filters, measured against the math

THE QUESTION

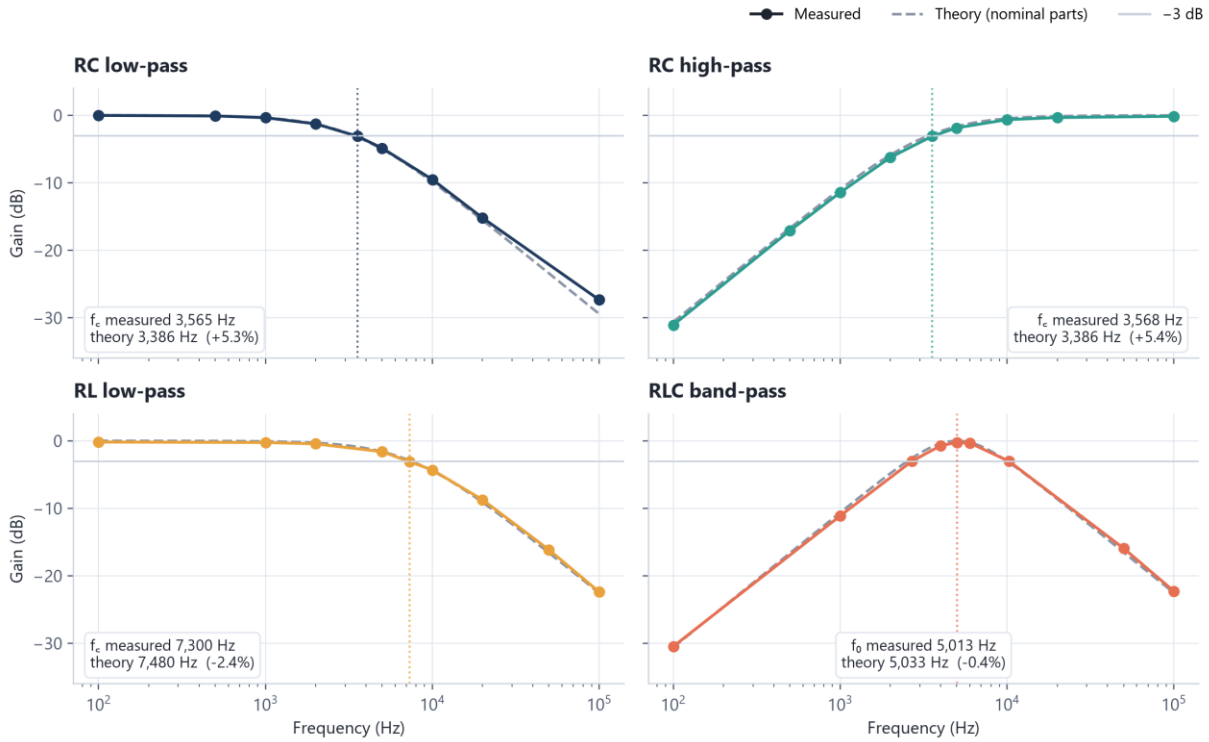
Do real RC, RL and RLC filters cut off where 1/(2πRC), R/(2πL) and 1/(2π√LC) say they should?

WHAT I DID

Swept each filter from 100 Hz to 100 kHz on the function generator and scope, and found each −3 dB point.

CONCLUSION

Every corner landed within 5.4 % of theory. Both RC cut-offs ran high by the same 5 %, consistent with the capacitor being about 95 nF (within its tolerance).



−0.4 %
RLC centre frequency vs theory (5,013 vs 5,033 Hz)

+5.3 %
Both RC cut-offs high: implies C ≈ 95 nF

Q 0.66
Band-pass sharpness, theory 0.67

Individual results sheet (Mar 2026). Measured points from my tables; theory uses the nominal 470 Ω, 100 nF, 10 mH.

Faraday's law, two ways

THE QUESTION

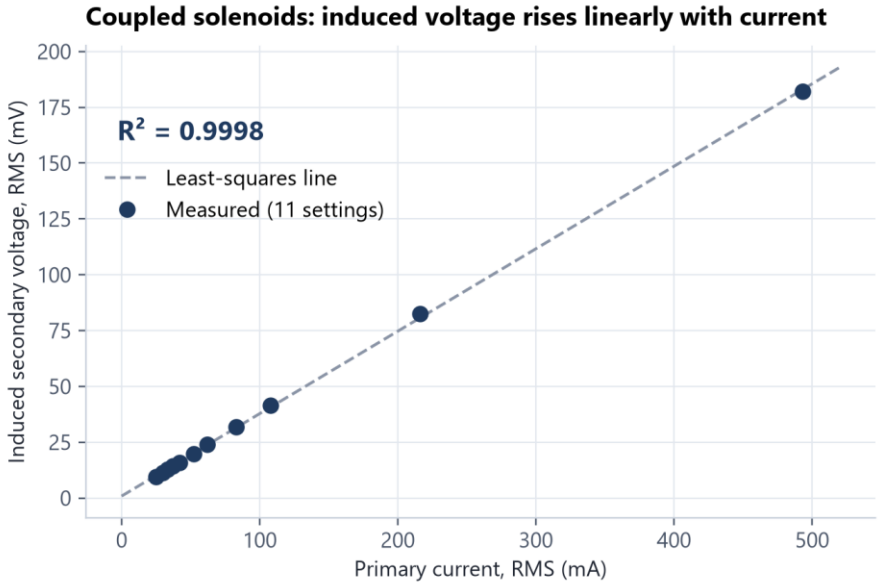
Is induced EMF proportional to the rate of change of flux, and can a flipped coil measure Earth's magnetic field?

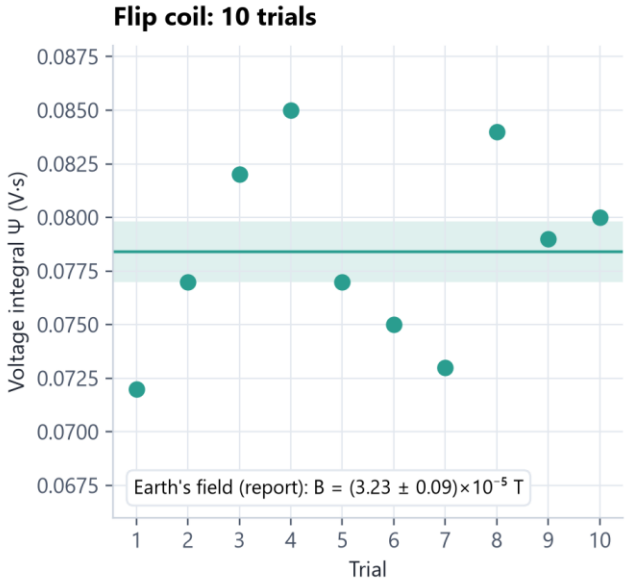
WHAT I DID

Ten 180° flips of a 100-turn coil through a ×200 amplifier; then 11 current settings on coupled solenoids, fit with LINEST in Excel.

CONCLUSION

Induced voltage is linear in current (R^2 = 0.9998). The flip coil gave B 32 % above the gaussmeter; likely local interference and coil alignment.





R^2 0.9998

Linear fit, induced EMF vs primary current

3.23 ± 0.09

Earth's field, $\times 10^{-5}$ T (fractional error propagation)

2.45

Gaussmeter reference, $\times 10^{-5}$ T

With lab partners Duong Hung Nguyen and James Pacia. Report written individually (Mar 2026).

When 90 Hz turns into 10 Hz

THE QUESTION

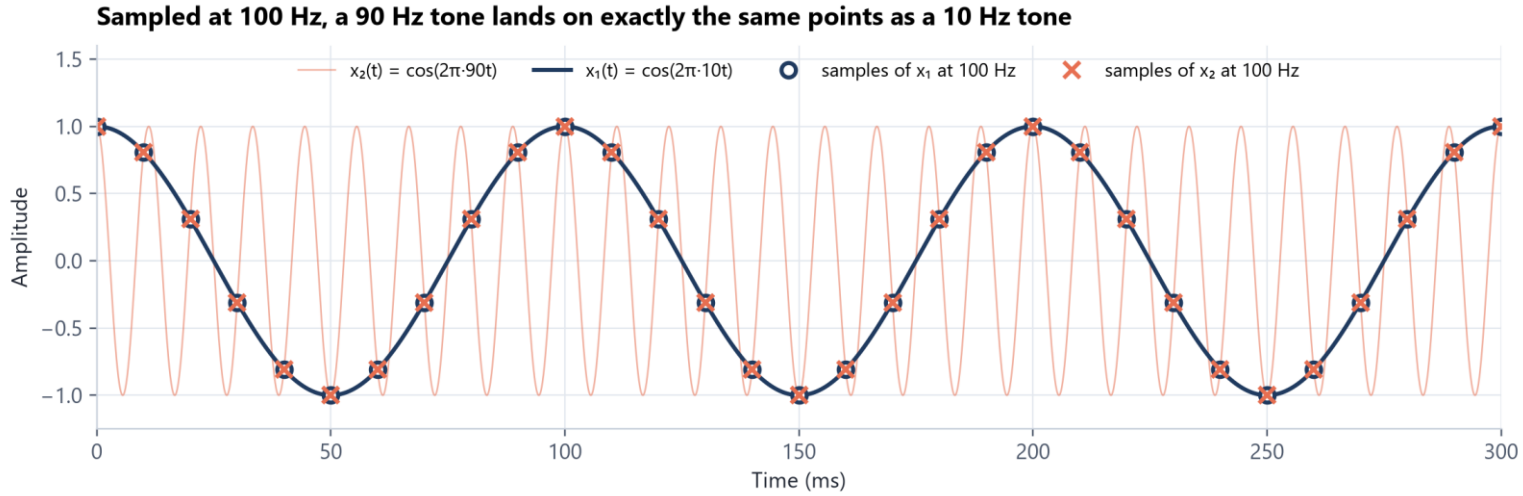
What happens when a signal is sampled slower than twice its highest frequency, in 1D and in images?

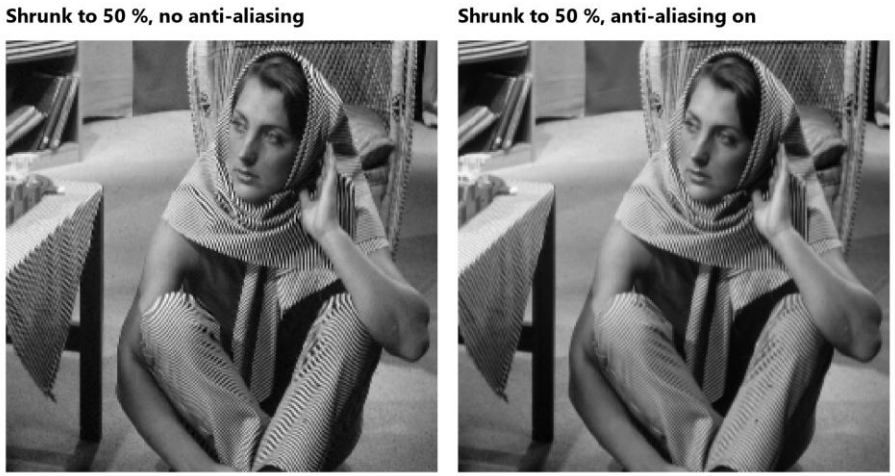
WHAT I DID

Sampled 10 Hz and 90 Hz cosines at 100 Hz and 1 kHz in MATLAB, then shrank a test image with anti-aliasing off and on.

CONCLUSION

At 100 Hz the two tones give identical samples (max difference 1.7×10^{-14}). Shrinking an image is resampling too: fine stripes alias into moiré unless low-passed first.





Left: stripes on the clothing turn into false wavy bands.

Right: filtered first, they fade to grey instead. Cost: a little blur.

WHAT THESE SHOW

Predict it, measure it, explain the gap.

Bench

Oscilloscope (AC/DC coupling), function generator, DMMs, breadboard builds

Simulation

LTspice: stepped parameters, .meas averages and ripple

Analysis

Excel LINEST, error propagation, MATLAB conv, sampling, imresize

Reporting

Spec checks, honest failures and their causes, Word reports

Data and every calculation: Lab_Highlights_Data.xlsx · anmol-portfolio-omega.vercel.app/docs